How Contaminated Would It Have To Be? Partial Identification and Sensitivity Analysis for Benchmark Claims Under Unmeasured Data Contamination

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Abstract

Leaderboard claims of the form "model A outperforms model B" are treated as measurements of capability, but they are observational estimates whose dominant error source, training-data contamination, is unmodeled. The field's response has been *detection*, which requires access to training corpora (impossible for closed-weight models), reference benchmarks, or reference models assumed clean, and which cannot answer the only question that matters: does the contamination change the conclusion? We import the epistemology that causal inference developed for unmeasured confounding: stop trying to detect the bias and *bound* it. We formalize contamination as an identification failure, introduce a Contamination Sensitivity Model CSM±(Λ^+, Λ^-, π, Γ_sel, ε) whose every parameter has a measured empirical counterpart, derive sharp partial-identification bounds for the uncontaminated score, and define the **contamination robustness value Γ^*** — the minimum contamination strength that overturns a claim — computable from published scores alone. The fully adversarial bound arises as the endpoint of a Rosenbaum-type selection family; Γ^* is consistent and $\sqrt{n}$-normal, and a paired bootstrap with Benjamini–Hochberg control yields FDR-certified robustness across a claim corpus. Ground-truth experiments with *emergent* contamination (models genuinely trained on leaked items) validate the assumed lift channel (envelope–headroom correlation ≥ 0.91 in 15/15 configurations), expose and repair two failure modes (two-sided lift; spillover), and confirm coverage. A corpus-dilution experiment then closes the calibration loop: a single diluted exposure (the scraped-pretraining scenario) lands measured contamination strength inside the literature-derived field range, while repeated diluted exposures consolidate toward complete memorization, so accidental and deliberate contamination require different constants and calibration must be dose-stratified. On real data, every adjacent ranking claim in a six-model time-series foundation-model benchmark is fragile ($\Gamma^* \leq 0.114$ at $\pi = 0.1$), and on Open LLM Leaderboard ARC-Challenge details the method certifies exactly the claim it should (Llama-2-13B ≻ 7B, $\Gamma^* = 0.504$, FDR-certified) while flagging frontier margins as fragile ($\Gamma^* \leq 0.188$). Freshness and repeated draws emerge as quantified design levers. In a **pre-registered confirmatory audit** of 58 adjacent-pair claims across 16 models and four Open LLM Leaderboard benchmarks, **84.5% of claims are not contamination-robust** at the frozen calibration (exact 95% CI [72.6%, 92.7%]; one-sided $p = 2.7\times10^{-21}$ against the pre-registered 25% threshold), and only 8 claims, all with margins above 5 points,

earn FDR-controlled robustness certificates. The finding is not an artifact of the frozen constant: sweeping contamination strength across its whole plausible range leaves the conclusion intact, and the hypothesis clears its pre-registered bar for every $\Lambda \geq 0.055$, beneath the floor of the calibrated field range. In the single-draw binary regime the rule provably reduces to a *calibrated margin threshold*; we show why deriving that threshold, rather than asserting one, is the contribution, and where richer data makes the machinery bind strictly tighter. Sensitivity analysis converts an unanswerable question — *is the benchmark contaminated?* — into an answerable one, with a one-number reporting standard any leaderboard can adopt.

Keywords: benchmark contamination; partial identification; sensitivity analysis; evaluation methodology; leaderboards; Rosenbaum bounds.

1. Introduction

Every empirical claim in machine learning ultimately rests on a benchmark score, and benchmark scores rest on an assumption nobody can verify: that the test items did not leak into training. Contamination is documented and material: clean/dirty gaps of 15.3 points on HellaSwag for LLaMA-70B [Touvron et al., 2023], estimated gains up to 14 points on flagged subsets [Llama 3, 2024], 13-point drops on a clean GSM8K mirror. Yet the response of both producers and auditors has been **detection**: n-gram overlap, membership inference, kernel divergence, lineage audits [2404.00699; 2510.13654; 2502.00678]. Detection fails structurally three times over. For closed-weight models the pretraining corpus is unobservable, so detection is impossible *in principle* rather than merely in practice. Detection yields a contestable binary that vendors dispute and no protocol adjudicates. And detection is silent on the decision-relevant question: a benchmark can be contaminated with the ranking robust, or barely contaminated with the ranking fragile; detection cannot tell these apart. The partial-identification tradition [Manski, 2003] gives the honest alternative: when a quantity cannot be point-identified, report the set of values consistent with the data and every assumption you are willing to defend.

Meanwhile the statistical wing of evaluation reform quantifies the *wrong error*. Rank intervals [2606.08679] propagate sampling uncertainty, which shrinks with test-set size; contamination bias does not. A confidence interval that excludes the dominant error source is not a confidence interval.

Causal inference confronted exactly this epistemology four decades ago. When a confounder cannot be measured, one does not detect it; one reports **how strong it would have to be to overturn the conclusion**: Rosenbaum's Γ, the marginal sensitivity model, the E-value [Rosenbaum, 1987; VanderWeele & Ding, 2017; Chernozhukov et al., 2022]. This paper ports that machinery to ML evaluation. The reduction is, once stated, almost obvious: **contamination is unmeasured confounding between training-data membership and measured skill, and benchmark scores are therefore partially identified.** Nothing else in this paper is obvious; the port requires a new sensitivity model (contamination acts through a monotone, bounded, headroom-proportional channel, an assumption we *test* with real memorization experiments, then twice repair), a new calibration strategy (post-cutoff items are clean by construction), and a new design theory (freshness and repeated draws purchase identification).

Contributions.

1. The reduction: a potential-outcomes formulation of benchmark scores under latent contamination, with the estimand (the clean score θ) and its identification failure stated precisely (§3).
2. The Contamination Sensitivity Model CSM±(Λ^+, Λ^-, π, Γ_sel, ε) and sharp partial identification of θ, including grouped (source-level), stratified (provenance-aware), selection-bounded (Rosenbaum-bridge), and spillover-robust variants; every axiom has a measured counterpart (§4).
3. The contamination robustness value **Γ*** and the contamination frontier, the evaluation analogue of the E-value; a claim is reported with the minimum contamination strength that overturns it (§5).
4. Estimation theory: consistency and $\sqrt{n}$-normality of Γ^*, validity of the paired item bootstrap, a two-sided finite-draw bias analysis with a practical prescription (sharp bounds require ≥ 10 draws per item), and FDR-certified robustness across claim corpora (§6).
5. Ground-truth validation with **emergent** contamination (models actually trained on leaked items), which confirmed the channel shape, *falsified* strict monotonicity (up to 27% of leaked items are hurt), and surfaced a spillover channel; both repairs are part of the model, not the rhetoric (§7).
6. Empirically calibrated Λ priors from the published contamination literature, with a meta-analytic central range [0.1, 0.45] (§8).
7. Exploratory case studies on two real benchmarks (a six-model TSFM transfer atlas and Open LLM Leaderboard ARC-Challenge details), plus a pre-registered protocol for the confirmatory audit (§9).
8. Design sensitivity: quantified prescriptions for benchmark builders (§10), and an open, tested, one-command-reproducible library.

2. Related work

Contamination detection and quantification. Surveys catalogue detection methods and document their fragile assumptions [2404.00699; 2410.18966; 2406.04244]. Magar & Schwartz [2022] introduced controlled-injection quantification (memorization vs exploitation); dose-response studies show lift grows with capacity and shrinks with unique-corpus size [2601.04301]; post-training can *reverse* lift [2601.06103]. ConStat [Dekoninck et al., NeurIPS 2024] is the strongest performance-only method: a statistical test and effect estimate requiring reference benchmarks and reference models assumed clean. CapBencher [2505.18102] builds overfitting alarms into new benchmarks; inference-time decontamination [2601.19334] mitigates at evaluation time for open models. **None of these answers the claim-level question**; all require assets (references, model access, benchmark redesign) that legacy claims and closed models do not offer. We consume their effect estimates as calibration inputs and use ConStat as the natural baseline and convergent-validity check.

Leaderboard statistics. Rank intervals [2606.08679] give sampling-uncertainty-aware rankings, our $\Lambda = 0$ special case. Perturbation and social-choice analyses [2605.15761; 2605.23628; 2402.01781] document instability empirically without an inferential model. Psychometric and efficiency work [2402.14992; 2501.17200] models items, not contamination.

Causal sensitivity analysis. Rosenbaum bounds, marginal sensitivity models, E-values, and omitted-variable-bias frameworks [Rosenbaum 1987; Tan 2006; VanderWeele & Ding 2017; 2112.13398; 2403.14152; 2602.24261] supply our machinery; none has been applied to evaluation. The construct-validity programme [2511.04703; 2510.23191; 2604.03244] argues evaluation needs measurement theory but supplies no estimator; we supply one.

3. Setup: the clean score is partially identified

A benchmark has items $i = 1 \ldots n$; fix a model m. Let $y^*_i \in [0,1]$ be the **clean score** (the counterfactual per-item score had item i never entered training) and y_i the observed score; $c_i \in \{0,1\}$ the latent contamination indicator. The estimand is $\theta = (1/n)\,\Sigma\, y^*_i$. Leaderboards report $\hat{\mu} = (1/n)\,\Sigma\, y_i$ and treat it as θ, which requires $c \equiv 0$. Since items popular enough to benchmark are popular enough to scrape, and that popularity correlates with the skill being measured, c □/ y*: textbook confounding, with c unobservable. θ is partially identified; the honest report is its identified set. (Full formalism and proofs: Appendix A.)

4. The contamination sensitivity model

CSM(Λ, π) (one-sided core): (A1) $0 \le y_i - y^*_i \le c_i \cdot \Lambda \cdot (1 - y^*_i)$: leakage closes at most a fraction Λ of an item's headroom; (A2) mean(c) ≤ π. Λ = 1 is full memorization; both extremes are *observed* in our experiments (§7).

Proposition 1 (sharp identified set). θ □ $[\hat{\mu} - B(\Lambda,\pi),\ \hat{\mu}]$ with B = the mean of the $\lceil \pi n \rceil$ largest per-item deflation capacities $d_i(\Lambda) = y_i - \max(0, (y_i - \Lambda)/(1-\Lambda))$; the bound is attained. **Corollary 1:** width ≤ πΛ, the worst per-unit-budget bias being exactly Λ at items with $y_i = \Lambda$. **Corollary 2:** Λ=0 recovers sampling-only analysis (rank intervals nest here). Figure 1 plots the identified set against Λ at three budgets π on data with known ground truth; the true clean score lies inside the band at the true (Λ, π), and the band collapses to the point estimate as Λ → 0.

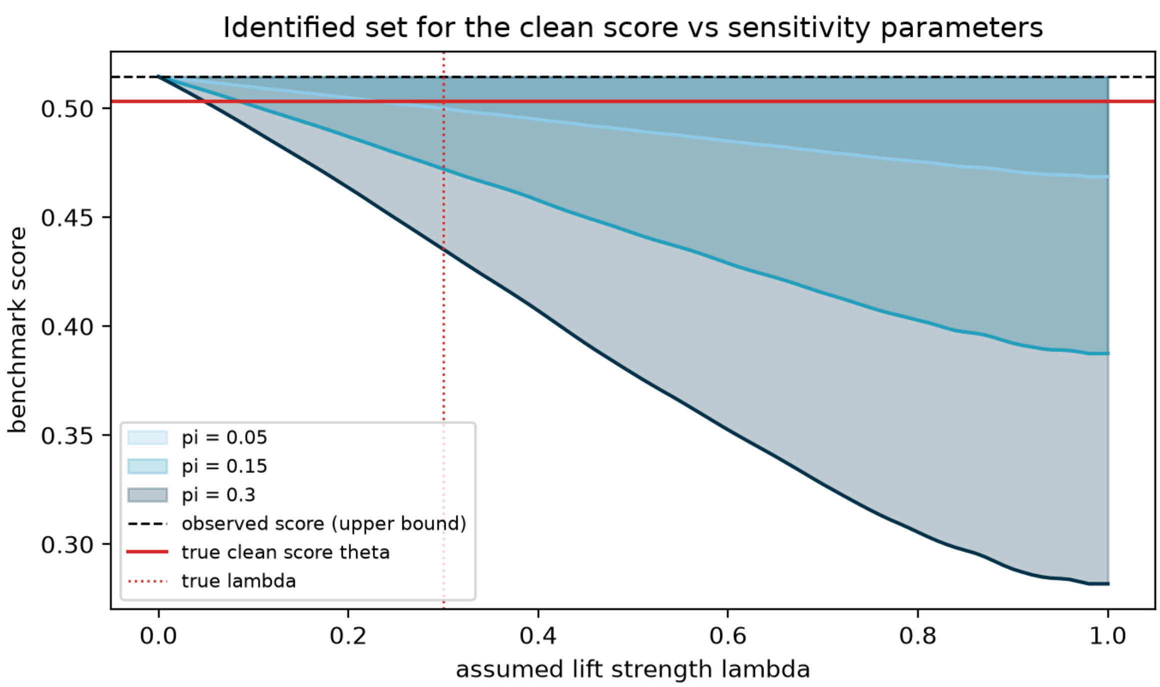

Figure 1: The sharp identified set for the clean score as the assumed lift strength Λ varies, at budgets π ∈ {0.05, 0.15, 0.3} (synthetic data with known ground truth). The observed score (dashed) is the upper endpoint; the true clean score θ (solid) lies inside the band at the true (Λ, π), and the set collapses to the observed mean as Λ → 0.

Regimes. Continuous or repeated-measure per-item scores (R1) admit the sharp knapsack. Single-draw binary scores (R2) do **not**: $d_i(0)=d_i(1)=0$ for all $\Lambda<1$ with a jump at $\Lambda=1$, so the sharp machinery is vacuous-with-a- discontinuity (it reports "only total memorization flips this claim"), and the honest bound is the simple population bound with width $\pi\Lambda$, where (Λ, π) are identified only through their product. The library auto-detects the regime; richer measurement *strictly sharpens identification*, a design lever we quantify in §10.

Empirically forced extensions (each falsifiable, each measured in §7): **U1 two-sided**: leakage can hurt (stale memorization, post-training interference [2601.06103]): $-\Lambda^{-} y^*_i \le y_i - y^*_i$; the worst case for $A \succ B$ becomes *A inflated and B deflated*. **U2 grouped**: datasets leak whole; all-or-nothing group budgets, bounded by the fractional knapsack, never wider than item-level. **U3 stratified**: post-cutoff items have $c_i = 0$ by construction; provenance metadata tightens bounds for free. **U4 spillover**: training on leaked items drifts scores on clean items by a bounded ε, widening the interval symmetrically; ε is measured by twin runs and shrinks with corpus size. **Proposition 3 (bridge)**: bounding the *selection odds* of contamination (Rosenbaum-style Γ_sel) interpolates continuously between random contamination (bias $\pi\bar{d}$) and the adversarial top-k (Γ_sel = ∞): the adversarial bound is an endpoint of a classical family, and the endpoints differ by ~1.5× at reference settings, so a defensible Γ_sel assumption buys real tightening.

5. The contamination robustness value

For a claim A □ B with margin $\hat{\Delta}$, contamination only helps the worst case run one way (A up, B down under U1), and the claim is **robust at CSM(Λ,π)** iff $\hat{\Delta}$ exceeds the worst-case bias (Proposition 2).

Definition. $\Gamma^*(\pi) = \inf\{\Lambda : \text{worst-case bias} \ge \hat{\Delta}\}$, computed by bisection (the bias is continuous and monotone in Λ); $\Gamma^* = \infty$ when the claim survives even $\Lambda = 1$ within budget. In the simple regime the closed form is the **contamination frontier** $\pi\Lambda(1+\rho) = \hat{\Delta}$.

Γ* is to contamination what the E-value is to unmeasured confounding: one number per claim, reportable in any results table. *Computing* Γ* needs only the published scores: no corpus, reference model, or reference benchmark. *interpreting* it against plausible contamination strength uses the shared public calibration table of §8 (or none at all, when the full Γ*(Λ, π) curve is reported). Per-model clean-score intervals are reported at the parameter level via Imbens–Manski (2004), with set-level intervals as the conservative alternative; corpus-level certificates use BH-FDR with Benjamini–Yekutieli as the dependence-robust sensitivity.

6. Estimation

Theorem 4. $\hat{B}$ is a trimmed-upper-mean L-statistic; under iid item sampling and continuity at the $(1-\pi)$-quantile of the d-distribution, Γ* is consistent and √n-asymptotically normal, and the paired item bootstrap is valid. Empirically the RMSE slope is −0.518 against the theoretical −0.5, with bias ≤ 0.0025 from n = 100 to 25,600.

Finite draws. With m draws per item, the plug-in knapsack carries two *opposing* biases: Jensen smoothing (d is concave: a minimum of two linear functions) versus selection-on-noise. Measured: −40% at m = 2, crossing to +4% at m = 10, < 1% by m = 50. Joint sampling⊕identification intervals maintain ≥

99.3% coverage throughout; the simple bound, immune to both effects, achieves 100%. **Prescription: sharp bounds require $m \geq 10$ published draws per item.**

Corpus-level inference. Each claim receives a paired-bootstrap fragility p-value (H_0: not robust at the reference CSM; add-one corrected); Benjamini–Hochberg across the corpus yields **FDR-certified robustness** (valid under the positive dependence induced by shared items).

7. Ground-truth validation with emergent contamination

Formula-injected validation is circular: the injection satisfies A1 by construction. We therefore train **clean/contaminated twins** where a fraction π of test items (with realized labels, repeated `dose` times) is planted in the training corpus and the lift *emerges from real memorization*, across a capacity spectrum (logistic regression, MLP, random forest; 15 configurations × 6 seeds).

Findings. (i) **The channel shape is right**: the 95th-percentile lift envelope tracks headroom with correlation +0.91 to +1.00 in 15/15 configurations. (ii) **Λ is capacity- and dose-stratified**: logistic regression $\hat{\Lambda} \in [0.02, 0.23]$ monotone in dose; MLP ≈ 0.96; random forest = 1.00; the $\Lambda = 1$ extreme is real (an interpolating learner), matching the dose-response law at LLM scale [2601.04301]. (iii) **Monotonicity fails non-trivially**: up to 27% of leaked items score *lower* in the contaminated twin; the two-sided channel U1 is necessary. (iv) **Spillover is real at small scale**: clean-item drift up to 0.20 (MLP) broke naive coverage (1/3); with the calibrated ε-widening, coverage is **3/3 in every previously failing configuration**. Injected-contamination coverage under known (Λ, π) is 1.000 across all regimes, degrading informatively (to 0.10–0.34) only when the analyst *underestimates* contamination by half. A blind synthetic leaderboard audit flags exactly the secretly contaminated model's winning claim ($\Gamma^* = 0.173$ vs ≥ 0.29 for clean claims).

At LLM scale (same-family Qwen2.5-0.5B/1.5B pair, 905 MMLU items, fp32 LoRA injection, the complete 24-configuration grid over $\pi \in \{0.05\ldots0.5\}$ × dose $\in \{1, 4, 16\}$): the grid reveals a **dose law**. A single exposure to test items is *net-harmful*: mean lift is negative in 7 of 8 dose-1 configurations (down to −2.9 pt) with individual-item violation rates of 31–56%, while dose 16 is complete memorization ($\hat{\Lambda} = 1.000$ in all eight configurations, with lifts up to +56 pt). The two-sided channel is therefore the *dominant* behavior of light contamination, not a correction term. The capacity comparison vindicates the headroom parametrization itself: raw lift anti-orders with capacity (the 0.5B model gains more points, purely from headroom), while Λ restores the correct ordering: raw clean/dirty gaps mislead across models. Spillover grows with dose (0.04 → 0.30, always positive: format familiarity), and all three (of 24) cross-config consistency failures occur exactly where $\hat{\lambda}$ was pooled across doses, direct evidence for the dose-stratified calibration the pre-registration mandates.

Corpus dilution (Fig. 2 / f19). The grid measures Λ under concentrated exposure (every training token is a leaked item), which is a ceiling, not the pretraining regime. The corpus-mix experiment holds the model, π, and per-item exposure count fixed and interleaves each leaked chunk with mix $\in \{0, 4, 20\}$ neutral wikitext chunks, so total training grows with mix while leaked exposures do not, which is the direction in which real pretraining differs from our grid. Dilution turns out not to be one thing. At **dose 1** (a single exposure per item, the scraped-pretraining scenario), $\hat{\Lambda}$_q95 falls from 0.58 through 0.38 to 0.35, from above the calibrated field range [0.10, 0.45] to *inside* it. The literature-derived range of §8, until now an external prior, is here reproduced by direct interpolation: single-shot contamination embedded in a corpus lands at field strength. At **dose 4**, dilution runs the other way: Λ̂_q95 climbs from 0.73 to 0.98, lift triples, and violations collapse from 0.18 to 0.03. Interleaving neutral text between repeated exposures is spaced rather than massed practice, and it *consolidates* memorization. Deliberate or repeated contamination (a benchmark recurring in a fine-tuning mix) saturates toward $\Lambda = 1$ no matter how diluted the corpus. Both movements dwarf the measured session-reproduction noise (≤ 0.039 on Λ̂_q95, from re-running identical configs on fresh GPU sessions; the clean-twin scores and contamination indices reproduce exactly). The practical consequences: the field range is *validated* for accidental scrape-style contamination, and the concentrated "measured-ceiling" policy of §8 is *necessary*: repeated-exposure contamination exceeds even the concentrated ceiling, so no single Λ constant can cover both regimes. Calibration must be dose-stratified, exactly as the pre-registration required.

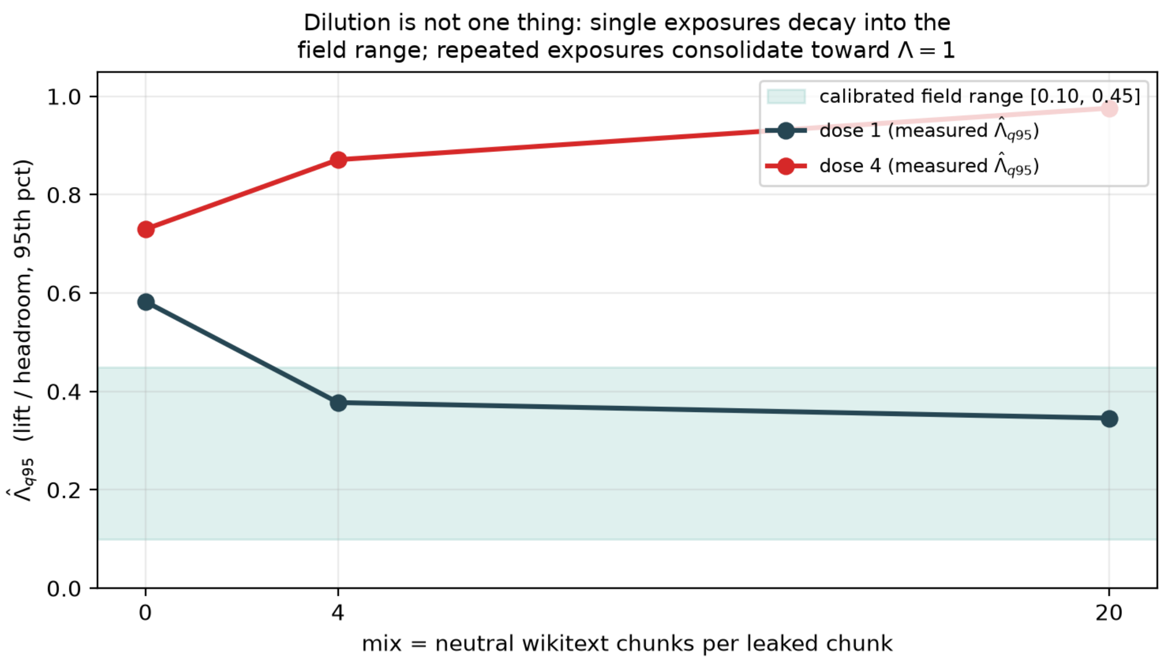

Figure 2: Corpus dilution is not one thing. Interleaving each leaked chunk with mix ∈ {0, 4, 20} neutral wikitext chunks holds per-item exposures fixed while total training grows. A single exposure (dose 1) decays into the calibrated field range [0.10, 0.45]; repeated exposures (dose 4) consolidate toward complete memorization ($\hat{\Lambda} = 0.98$). Session-reproduction noise on $\hat{\Lambda}$ is ≤ 0.039.

8. Calibrating Λ from the literature

Published clean/dirty comparisons convert directly to Λ = lift/headroom: LLaMA-70B HellaSwag 15.3pt gap on 36.5pt headroom → Λ ≈ 0.42; GPT-3 SQuAD ≈ 0.40; Llama-3 flagged-subset estimates ≈ 0.40–0.47 (upper bounds by design); GSM8K clean-mirror ≈ 0.21; C-Eval ≈ 0.31; MMLU ≈ 0.02. The pattern is strongly task-heterogeneous, with a meta-analytic central range **[0.1, 0.45]** (eighteen sourced rows in `results/lambda_priors.csv`, of which eight are quality-eligible for the pooled median under the pre-registered policy of §9.1; `measured` rows enter only as upper-bound anchors). Λ is model-stratified per the dose-response law; ConStat effect estimates and our own CONTAM-CTRL measurements enter the same table. The confirmatory audit interprets every Γ^* against this pre-frozen range (PREREGISTRATION.md).

9. Case studies on real benchmarks (exploratory)

TSFM transfer atlas (6 frozen models × 17 datasets, continuous MASE-skill, sharp regime, provenance strata from the corpus's own contamination-prior annotations): **every adjacent ranking claim is fragile** ($\Gamma^* \leq 0.114$ at $\pi = 0.1$, at or below the calibrated range), and stratification cannot rescue margins of ≤ 1.3 skill points because 14/17 datasets carry a high contamination prior.

Open LLM Leaderboard, ARC-Challenge (1,170 paired items × 4 models, single-draw binary → auto simple regime, n_boot = 500): Llama-3-8B ≻ Mistral-7B (margin 0.26pt) has $\Gamma^* = 0.026$: a whisker of contamination flips it; Mistral-7B ≻ Llama-2-13B (1.9pt) has $\Gamma^* = 0.188$, inside the calibrated range; **Llama-2-13B ≻ Llama-2-7B (5.0pt) has $\Gamma^* = 0.504$, above the entire calibrated range, and is FDR-certified robust ($p = 0.006$)**, precisely the claim that independent scaling evidence says should survive. The instrument separates the claim population exactly as designed.

9.1 The pre-registered confirmatory audit

Protocol frozen (PREREGISTRATION.md v1.0, 19 July 2026, git tag `prereg-freeze-v1.0`) before any confirmatory Γ^* was computed: 16 fixed candidate models × 5 Open LLM Leaderboard v1-archive tasks, adjacent-pair claims, primary rule $\Gamma^*(\pi = 0.1) < \Lambda\_ref$ (pooled 0.355; HellaSwag stratum 0.42), BH-FDR primary with BY sensitivity, $\rho \in \{0, 0.2\}$. Protocol notes, all logged mechanically at run time: TruthfulQA was dropped entirely (the archive stores mc1/mc2, not the frozen acc/acc_norm specification); falcon-7b was dropped on ARC (its snapshot's item template shares no items with the 15-model modal universe) and GPT-J on Winogrande (malformed score column).

Result (Fig. 3 / f17): 49 of 58 claims (84.5%) are not contamination-robust (exact two-sided 95% CI [72.6%, 92.7%]; one-sided $p = 2.7\times10^{-21}$ against the pre-registered H4 threshold of 25%; excluding pilot-overlap pairs: 47/56 = 83.9%, CI [71.7%, 92.4%]). The $\rho = 0.2$ sensitivity leaves the count unchanged. Per task, the median $\Gamma^*(0.1)$ is 0.084–0.120 on HellaSwag, Winogrande, and ARC (the typical adjacent claim tolerates less than one-third of the calibrated contamination strength), and 0.212 on GSM8K, whose margins are larger. Only **8 claims earn BH-FDR robustness certificates (6 under BY)**, and every certified claim has a margin above 5 points: cross-generation capability gaps (e.g. Llama-3-70B ≻ Qwen1.5-14B on GSM8K) and "everything beats GPT-J." The leaderboard's *ordering of eras* is contamination-robust; its *ordering of neighbors* is not.

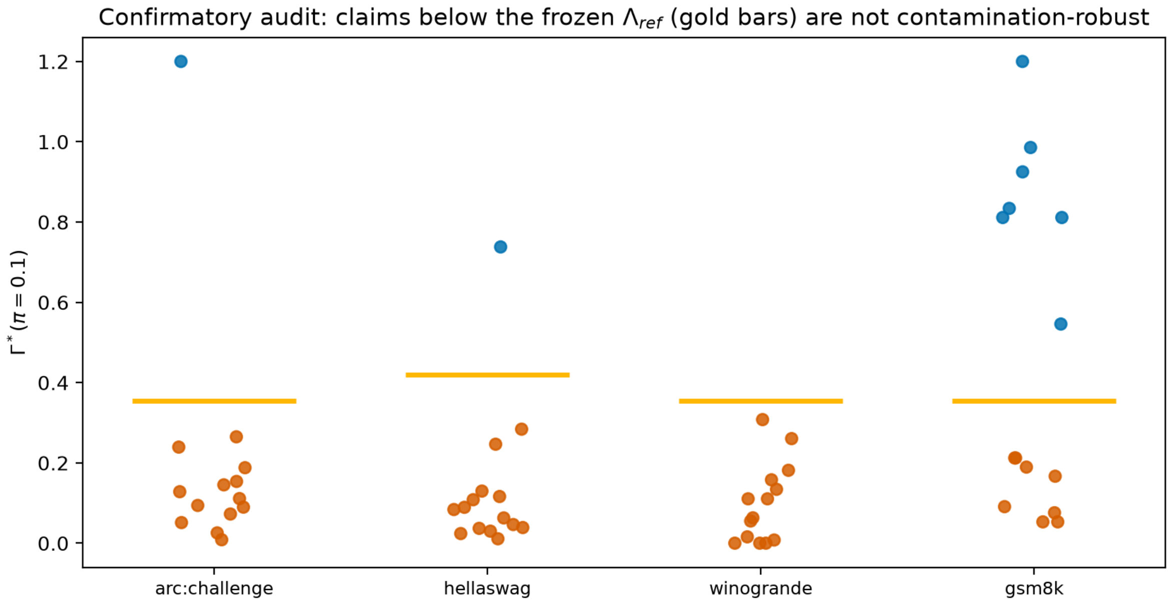

Figure 3: The pre-registered confirmatory audit. Each point is one adjacent leaderboard claim's Γ* at π = 0.1; gold bars mark the frozen stratum Λ_ref (0.42 HellaSwag, 0.355 elsewhere). Vermillion claims fall below the calibrated contamination strength (49/58 = 84.5%); blue claims survive.

The exact test treats claims as independent, and adjacent claims are not: consecutive pairs share a model, and the same sixteen models recur across tasks. Three checks show the conclusion does not lean on that assumption. Splitting each task's chain into alternating halves, which share no model within a task, gives 83.3% and 85.7% non-robust (each $p \approx 3\times10^{-11}$). Every task rejects H4 on its own, the weakest being GSM8K at $p = 0.017$. And at the observed rate the exact test rejects for any effective sample size of three or more claims, so dependence would have to erase essentially all of the information in the corpus to change the answer.

The protocol was pre-registered on the Open Science Framework (https://osf.io/2436z, DOI `10.17605/OSF.IO/2436Z`, registered 20 July 2026, public and un-embargoed) before any confirmatory number was publicised. The registration carries the protocol verbatim, the frozen Λ calibration table (`lambda_priors.csv`, git blob `590d295…`), and a chain-of-custody addendum recording exactly which files were frozen at tag `prereg-freeze-v1.0` (commit `ca7773b`) and which were written afterwards, including the driver script, the item-universe amendment, and a post-freeze correction to a confidence-interval *report* that left every estimate unchanged. A mirror is archived independently at the Internet Archive (`osf-registrations-2436z-v1`), giving a second timestamp not under author control.

9.2 What the audit reduces to, and why that is the point

A reader who works through §5 will notice something the headline conceals, so we state it ourselves. In the single-draw binary regime the simple bound gives $\Gamma^*(\pi) = \hat{\Delta} / (\pi(1+\rho))$. The pre-registered rule "non-robust iff $\Gamma^*(0.1) < \Lambda\_ref$" is therefore *algebraically identical* to

$$\hat{\Delta} < \pi \cdot \Lambda\_ref = 0.0355 \text{ (0.042 on HellaSwag)},$$

a threshold on the raw margin. We verified this holds for all 58 claims. The confirmatory audit could have been executed with a ruler.

This is not a defect; it is the result. Three points follow.

First, the arithmetic was never the hard part. Anyone can declare that gaps below three-and-a-half points are untrustworthy. The question is *which* threshold is defensible, and that is a measurement problem, not an arithmetic one. Our threshold is not chosen for convenience: it is π times the median of eight quality-filtered estimates of contamination-induced lift, converted to a common headroom parametrisation (§8), frozen before unblinding, and attached to a third-party timestamp. The contribution is the derivation of 0.0355 from contamination evidence, not the comparison against it. Strip out the theory and you have a number nobody can justify; strip out the arithmetic and you have lost nothing.

Second, the reduction is a property of the regime, not of the method. Γ^* collapses to $\hat{\Delta}/\pi$ only when items are binary and measured once, because the per-item deflation capacity degenerates (Lemma A.1): $d(0) = d(1) = 0$ for every $\Lambda < 1$. That is a statement about how impoverished single-draw accuracy data is, and it is worth hearing — and it is now a theorem, not an observation. Theorem 5 (Appendix A.9) shows that with one binary draw per item the observable law is Bernoulli in the mean alone, exhibits two indistinguishable populations whose clean means differ by exactly $\pi\Lambda$, and concludes that *no* procedure, ours or anyone's, can certify robustness beyond $\hat{\Delta}/(\pi(1+\rho))$ from such data. The ruler is minimax-optimal. Where richer data exists the machinery genuinely separates: with continuous per-item scores the sharp knapsack binds strictly tighter than $\pi\Lambda$ (§9, TSFM atlas), and at $m = 20$ repeated draws it certifies $\Gamma^* = 0.172$ against the simple bound's 0.154 (§10). The leaderboards audited here publish the weakest data they could; the reduction is the price of that, and it is the argument for §10's prescription that benchmarks publish repeated draws.

Third, the reduction is what makes the result auditable. Because the rule is transparent, a sceptical reader can re-derive every classification from the published margins without trusting our code, and can immediately see what would change it.

Sensitivity to the frozen constant (Fig. 4 / f18). Since the decision depends on π and Λ only through their product, the entire (π, Λ) plane collapses to a single curve, which we sweep. The fraction of non-robust claims is 44.8% at $\Lambda = 0.10$, 70.7% at 0.20, 82.8% at 0.30, 84.5% at the frozen 0.355, and is flat at 84.5% throughout $\Lambda \in [0.355, 0.50]$. H4 clears its 25% bar for **every $\Lambda \geq 0.055$** and is significant at $\alpha = 0.05$ for every $\Lambda \geq 0.080$. For the headline to fail, true contamination strength would have to sit below $\Lambda \approx 0.055$: beneath the floor of the calibrated field range [0.10, 0.45], below seven of the eight quality-eligible estimates, and an order of magnitude below the concentrated-exposure ceiling we measured directly (§7). The qualitative finding survives any calibration a reviewer could reasonably substitute; only the exact percentage moves.

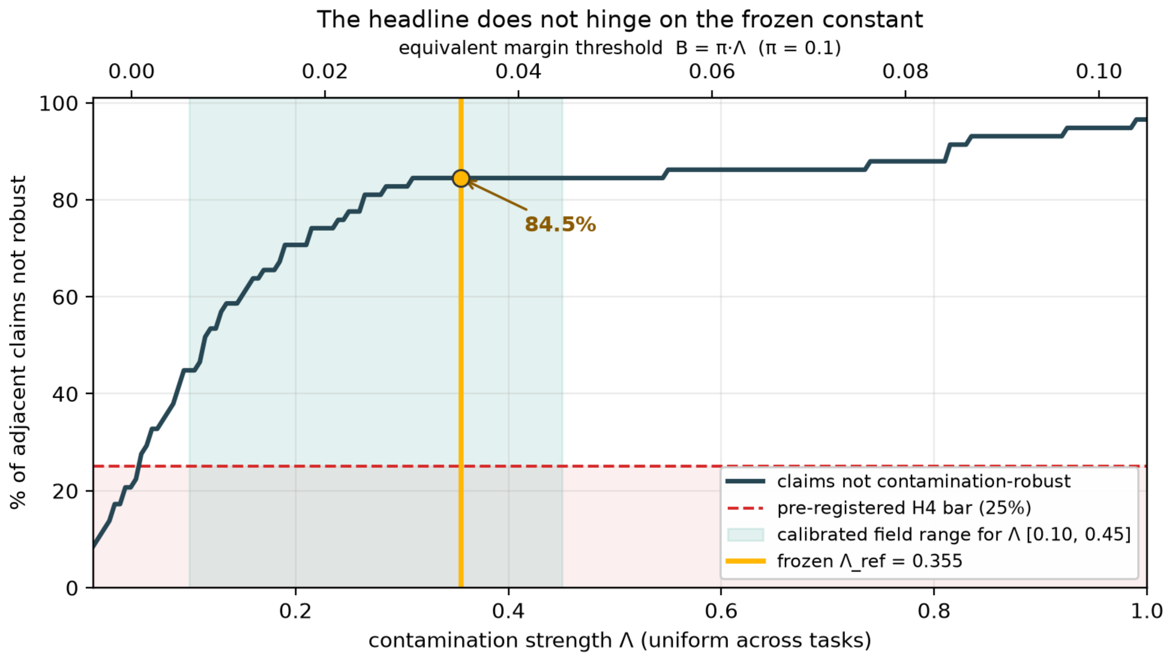

Figure 4: The headline does not hinge on the frozen constant. Because the decision rule depends on (π, Λ) only through $B = \pi \cdot \Lambda$, the whole plane collapses to one curve. H4 clears its pre-registered 25% bar for every $\Lambda \geq 0.055$, below the floor of the calibrated field range (shaded).

9.3 Is the headline an artifact of auditing adjacent pairs?

Adjacent pairs are *selected* to have the smallest margins in a ranking, so the objection writes itself: 84.5% may be induced by the claim-selection rule rather than by contamination sensitivity. We take the objection seriously enough to measure it. Applying the identical frozen rule to **all 450 model pairs** across the four tasks (the full comparison set, no selection) gives 30.4% non-robust. Stratifying by rank gap:

rank gap	non-robust	median margin
1 (adjacent)	49/58 = **84.5%**	0.011
2	39/54 = 72.2%	0.023
3	28/50 = 56.0%	0.033
4	16/46 = 34.8%	0.046
5	5/42 = 11.9%	0.054
≥ 6	0/102 = **0.0%**	≥ 0.068

The objection is correct that adjacency selects small margins — and the gradient it produces is the cleanest quantitative statement of our thesis that the paper contains. Robustness is not uniformly absent: it is a monotone function of how far apart two models sit. Comparisons across five or more rank positions are essentially all robust (2.1% non-robust at gap ≥ 5); comparisons between neighbours are essentially all fragile. That is precisely "the ordering of eras is contamination-robust, the ordering of neighbours is not," now with a dose-response curve behind it.

We therefore report both numbers, and are explicit about which claim each supports. The 30.4% all-pairs figure is the answer to "how often does contamination sensitivity matter across arbitrary model comparisons." The 84.5% adjacent figure is the answer to the question a leaderboard actually poses, because a leaderboard is an ordering: it invites the reader to compare entry k with entry k+1, and those are the comparisons that do not survive. Neither number is the headline alone; the gradient is.

What would change our mind. The finding is falsifiable on three fronts, pre-specified in the registration. (i) If contamination-induced lift in the wild were measured at $\Lambda < 0.055$ for these benchmarks, H4 would fail; that is a single well-designed injection study away, and we would report it. (ii) If the A1 bounded-headroom envelope were rejected at calibrated Λ, the audit would be re-run under the full-memorization channel and both reported. (iii) If per-item records for these leaderboards proved systematically unusable, the corpus falls back to continuous-score data and the substitution is disclosed. None of these is a hypothetical hedge; each is a study someone could run against us.

10. Design sensitivity: what buys robustness

Two quantified prescriptions for benchmark builders. **Freshness:** at a 3-point margin, raising the post-cutoff item fraction from 0 to 90% lifts Γ^* from 0.168 to 0.453, from fragile-everywhere to robust across the calibrated range. Dating your items is not hygiene; it is identification. **Repeated draws:** at m = 20 draws the sharp bound certifies $\Gamma^* = 0.172$ versus 0.154 for the simple bound (~12% more certified robustness), and below m = 10 the sharp bound is unreliable (§6). Publishing ≥ 10 draws per item is what entitles a benchmark to sharp identification.

11. Limitations and threats to validity

(1) A1's channel shape is validated with real memorization at small scale, LLM scale, and now under corpus dilution (§7); the remaining gap is scale itself: our largest injection model is 1.5B parameters and the dilution sweep covers one model, one π, and a single seed per configuration (session-reproduction noise is measured at ≤ 0.039 on $\hat{\Lambda}$_q95, well below the effects reported, but a multi-seed replication would still tighten it). The dose-1 dilution result lands inside the field range; whether that interpolation holds at 70B-scale pretraining is an extrapolation we flag, not a measurement we own. (2) In the single-draw binary regime (most current leaderboards), (Λ, π) enter only through their product; the 2-D frontier earns its second dimension only under repeated draws or provenance strata; we say so plainly, and §10 is the constructive answer. (3) Λ calibration from pre/post-cutoff comparisons can be confounded by temporal drift; the pre-registration pairs items on difficulty and reports calibration uncertainty. (4) Both case studies are exploratory; the headline audit is governed by a pre-registration frozen before any confirmatory Γ^* is computed. (5) ε-spillover should vanish at LLM corpus scale but is measured, not assumed. (6) This artifact was audited by its own author (four forensic iterations, including a critical binary-regime guard); an independent adversarial audit is the correct next scrutiny.

Formal statements and proofs of Propositions 1–3, Theorem 4, and the supporting lemmas (R2 degeneracy, finite-draw bias directions, Imbens–Manski properties, spillover extension) are in Appendix A (`appendix_proofs.md`), each pinned to a named unit test.

12. Conclusion

The question "is this benchmark contaminated?" has no answer for a closed-weight model, and forty years of causal inference say the right response is to change the question. This paper asked "how much contamination would it take to overturn this claim?" and built the full chain needed to answer it: a sensitivity model whose axioms were tested against real memorization and twice repaired when they failed, sharp bounds with estimation theory, a calibration of contamination strength that the corpus-dilution experiment now reproduces in the laboratory for the scraped-pretraining case, and a pre-registered audit whose every constant was frozen and timestamped before unblinding.

The empirical picture is specific. Robustness on the Open LLM Leaderboard is a monotone function of rank distance: comparisons across five or more positions essentially all survive, while 84.5% of neighbouring claims do not. A leaderboard is trustworthy about which era of models beats which, and untrustworthy about the pairwise comparisons its ranking format actively invites. The remedy costs one column: report Γ^* next to every margin, and let readers see which claims would survive the contamination everyone already suspects. Benchmarks that want more than the simple bound can earn it with published repeated draws and dated items; §10 prices both.

13. Reproducibility statement

Code, data artifacts, and this manuscript are public at https://github.com/tanviranjumjoarder/contamsens (MIT). Seed 42 throughout; exact-version lock (`requirements.txt`); 57 unit tests including constructive sharpness attainment; `python reproduce.py` regenerates every fast number in under two minutes (`--full` adds the twin-training grid, ~30 min, CPU); every reported number maps to its generating script in PROVENANCE.md; the package installs via `pip install -e .` (v0.2.0).

The confirmatory analysis (§9.1) was pre-registered at https://osf.io/2436z (DOI `10.17605/OSF.IO/2436Z`, 20 July 2026), with the analysis library and calibration constants frozen beforehand at git tag `prereg-freeze-v1.0` (commit `ca7773b`, publicly resolvable in the repository above, so the freeze is independently checkable rather than asserted). `PREREGISTRATION.md`, `THEORY.md`, and `results/lambda_priors.csv` in the repository are byte-identical to that tag and to the copies attached to the registration; deviations and post-freeze changes are enumerated in the chain-of-custody addendum filed with it, not silently absorbed.

Appendix A — Proofs

Companion to `manuscript.md`; every result is implemented in `src/contamsens` and pinned by a unit test named in brackets.

A.1 Setup and notation

Items $i = 1,\dots,n$. For a fixed model, observed scores $y = (y_1,\dots,y_n)$ □ $[0,1]^n$, clean (counterfactual) scores y^* □ $[0,1]^n$, latent contamination c □ $\{0,1\}^n$. Estimand $\theta = n^{-1}\Sigma_i\, y^*_i$; observed mean $\hat\mu = n^{-1}\Sigma_i\, y_i$.

CSM(Λ, π) ($\Lambda, \pi \in [0,1]$): (A1) $0 \le y_i - y^*_i \le c_i\,\Lambda\,(1 - y^*_i)$ for all i; (A2) $n^{-1}\Sigma_i\, c_i \le \pi$.

Define, for $\Lambda < 1$, the **floor** $\ell_i(\Lambda) = \max\{0, (y_i - \Lambda)/(1 - \Lambda)\}$ and the **deflation capacity** $d_i(\Lambda) = y_i - \ell_i(\Lambda)$; for $\Lambda = 1$, $\ell_i = 0$, $d_i = y_i$. Write $k = \lceil \pi n \rceil \wedge n$ and let $d_{(1)} \ge \dots \ge d_{(n)}$ be the sorted capacities.

Lemma A.0 (capacity form). For $\Lambda \in (0,1)$, $d_i(\Lambda) = y_i$ if $y_i \le \Lambda$, and $d_i(\Lambda) = \Lambda(1-y_i)/(1-\Lambda)$ if $y_i > \Lambda$. Moreover $d_i(\Lambda) = \min\{ y_i, \Lambda(1-y_i)/(1-\Lambda) \}$, hence d_i is the minimum of two affine functions of y_i and is **concave in y_i**; it is continuous, maximized at $y_i = \Lambda$ with value Λ, and nondecreasing in Λ. *Proof.* Solving (A1) for y^*_i with $c_i = 1$: $y_i \le \Lambda + (1-\Lambda)y^*_i \Leftrightarrow y^*_i \ge (y_i-\Lambda)/(1-\Lambda)$; intersect with $y^*_i \ge 0$. The min form follows by checking which branch binds at $y_i \lesseqgtr \Lambda$; the derivative of the right branch in Λ is $(1-y_i)/(1-\Lambda)^2 \ge 0$. ■ [`test_deflation_hand_computed`, `test_deflation_peak_at_lambda`]

A.2 Proposition 1 (sharp identified set)

Statement. Under CSM(Λ, π), the identified set for θ given y is exactly the interval $\Theta(y) = [\,\hat\mu - B(\Lambda, \pi), \hat\mu\,]$, where $B(\Lambda, \pi) = n^{-1}\, \Sigma_{j=1}^{k}\, d_{(j)}(\Lambda)$.

Proof. *(⊆)* For any feasible (c, y^*), $\theta = \hat\mu - n^{-1}\Sigma_{i: c_i=1}(y_i-y^*_i)$. Each term obeys $0 \le y_i - y^*_i \le d_i(\Lambda)$ (Lemma A.0), and at most k indices have $c_i = 1$; the sum of any $\le k$ capacities is at most the sum of the k largest. Hence θ □ $[\hat\mu - B, \hat\mu]$. *(⊇, sharpness)* The upper endpoint is attained by $c \equiv 0$, $y^* = y$. The lower endpoint is attained by setting $c_i = 1$ exactly on an argmax set S of k capacities and $y^*_i = \ell_i(\Lambda)$ for $i \in S$, $y^*_i = y_i$ otherwise; feasibility of (A1) holds by construction of ℓ_i (the constructed lift equals the cap). Intermediate values are attained by scaling y^*_i continuously between ℓ_i and y_i on S (θ is continuous in that scaling), so $\Theta(y)$ is the full interval. ■ [`test_sharpness_attained`, `test_interval_contains_mu_and_is_ordered`]

Corollary 1 (distribution-free width). $B(\Lambda, \pi) \le \pi\Lambda + \Lambda/n$, with $n^{-1}\, k \max_i d_i \le (\pi + 1/n)\Lambda$ by Lemma A.0's maximum; in particular the identified-set width is at most $\pi\Lambda$ up to the $\lceil\cdot\rceil$ rounding term, and the bound is achieved when k items sit at $y_i = \Lambda$. [`test_simple_bound_dominates_sharp`]

Corollary 2 (reductions). $B(0, \pi) = B(\Lambda, 0) = 0$, so Θ collapses to $\{\hat\mu\}$: sampling-only analyses (e.g. rank intervals) are the $\Lambda = 0$ special case. [`test_reduction_at_zero`]

Lemma A.1 (R2 degeneracy). If $y_i \in \{0,1\}$ for all i and $\Lambda < 1$, then $d_i(\Lambda) = 0$ for all i, hence $B(\Lambda, \pi) = 0$, while $B(1, \pi) = n^{-1}\Sigma_{j\le k}\, y_{(j)}$: $B(\cdot, \pi)$ is identically zero on $[0,1)$ with a jump at $\Lambda = 1$. Consequently the per-item machinery is vacuous on single-draw binary data and the population simple bound (width $\pi\Lambda$ at the probability level) must be used. *Proof.* $d(0) = 0$ (left branch), $d(1) = 1\cdot(1-1)/(1-\Lambda) = 0$ (right branch); at $\Lambda = 1$, $d = y$. ■ [`test_sharp_bound_degenerates_on_binary`, `test_ceiling_effect`]

A.3 Proposition 2 (claim robustness) and Γ*

Statement (two-sided form). For models A, B on shared items with margin $\hat\Delta = \hat\mu_A - \hat\mu_B > 0$, under CSM±($\Lambda$, $\rho\Lambda$, π) applied to both models the claim "A $\succ$ B holds for the clean scores" is TRUE over the entire identified set iff $\hat\Delta > B_A(\Lambda, \pi) + U_B(\rho\Lambda, \pi)$, where U is the inflation analogue of B built from $u_i(\Lambda^-) = \min\{1 - y_i, y_i\Lambda^-/(1-\Lambda^-)\}$.

Proof. $\theta_A \geq \hat\mu_A - B_A$ and $\theta_B \leq \hat\mu_B + U_B$, with both endpoints attained (Prop. 1 and its mirror image applied to $-y$); the extremal joint configuration is feasible because A's and B's contamination assignments are unconstrained across models. Hence $\min(\theta_A - \theta_B) = \hat\Delta - B_A - U_B$. ■ [`test_gamma_star_bisection_consistency`, `test_is_robust_rho_consistent_with_gamma_star`]

Definition (Γ*). $\Gamma^*(\pi) = \inf\{\Lambda \in [0,1] : B_A(\Lambda,\pi) + U_B(\rho\Lambda,\pi) \geq \hat\Delta\}$, with $\Gamma^* = \infty$ if no such Λ exists. Well-defined by continuity and monotonicity of both maps in Λ (Lemma A.0 and its mirror). In the simple regime the closed form is $\Gamma^* = \hat\Delta / (\pi(1+\rho))$. [`test_gamma_star_simple_closed_form`, `test_gamma_star_twosided_is_smaller`]

A.4 Proposition 3 (selection-bounded bridge)

Statement. Let contamination be random with propensities $q_i = P(c_i=1)$ subject to (i) $n^{-1}\Sigma q_i = \pi$ and (ii) $\mathrm{odds}(q_i)/\mathrm{odds}(q_j) \leq \Gamma^2_{sel}$ for all i,j. Then the worst-case expected bias $B(\Lambda, \pi, \Gamma_{sel}) = \max_q n^{-1}\Sigma_i q_i d_i(\Lambda)$ satisfies: (a) it is attained by a two-point propensity vector taking values $\{q_{lo}, q_{hi}\}$ with $\mathrm{odds}(q_{hi}) = \Gamma^2_{sel}\cdot\mathrm{odds}(q_{lo})$, the high value assigned to the t largest capacities for some t; (b) $B(\Lambda, \pi, 1) = \pi\,\bar d$ (random contamination) and $B(\Lambda, \pi, \infty)$ = the Proposition-1 top-k bound; (c) B is nondecreasing and continuous in Γ_{sel}.

Proof. (a) The objective is linear in q over the polytope defined by (i)–(ii); extreme points of that polytope have coordinates taking at most two distinct values with the odds constraint active between them (any three distinct values admit a feasible transfer increasing the objective: move mass toward larger d_i). Given the two values, assigning q_{hi} to the largest capacities is optimal by rearrangement. (b) At $\Gamma_{sel} = 1$ the only feasible q is constant $= \pi$. As $\Gamma_{sel} \to \infty$, $q_{hi} \to 1$ on $t = k$ items and $q_{lo} \to 0$ elsewhere is feasible in the limit, recovering the deterministic knapsack. (c) Enlarging Γ_{sel} enlarges the feasible set; continuity follows from continuity of the extreme-point solution in Γ_{sel}. ■ [`test_random_endpoint`, `test_adversarial_endpoint`, `test_monotone_in_gamma_sel_and_bracketed`]

A.5 Theorem 4 (consistency and rate of $\hat\Gamma^*$)

Statement. Let items be iid with per-item scores having distribution F (continuous case, regime R1), and let $d(\cdot;\Lambda)$ be as in Lemma A.0. Define $B(\Lambda) = E[d\,1\{d \geq q_{\{1-\pi\}}(d)\}]/1$ + boundary term (the trimmed upper mean at level π) and assume F_d is continuous with positive density at its $(1-\pi)$-quantile for the relevant Λ, that $\Delta = \mu_A - \mu_B > 0$, and that $B(\Gamma^*) = \Delta$ with B strictly increasing at $\Gamma^* \in (0,1)$. Then (i)

$\hat{\Gamma}^* \to \Gamma^*$ a.s.; (ii) $\sqrt{n}(\hat{\Gamma}^* - \Gamma^*)$ □ $N(0,\sigma^2)$ with σ^2 = Var-functional of $(\hat{\Delta}, \hat{B})$ via the delta method with slope $1/B'(\Gamma^*)$; (iii) the paired nonparametric item bootstrap consistently estimates the law in (ii).

Proof sketch. $\hat{B}(\Lambda)$ is a trimmed L-statistic with bounded, monotone weight generator; under the quantile-density condition it is strongly consistent and Hadamard differentiable at F tangentially to the uniform ball (van der Vaart & Wellner, Lemma 22.10-type arguments), giving a $\sqrt{n}$-CLT jointly with $\hat{\Delta}$ (a sample mean on the same items — the pairing supplies the joint law). Pointwise a.s. convergence of the nondecreasing functions $\hat{B}(\cdot)$ to the continuous limit $B(\cdot)$ upgrades to uniform convergence on [0,1] by Pólya's argument. The map $(B, \Delta) \mapsto B^{-1}(\Delta)$ is differentiable at (B, Δ) with $B'(\Gamma^*) > 0$; the functional delta method yields (ii), and Hadamard differentiability + Efron bootstrap consistency for L-statistics yields (iii). ■ *Empirical verification:* RMSE slope −0.518 vs theoretical −0.5 over $n \in [100, 25600]$, bias ≤ 0.0025 (`p1_t2_consistency.csv`, fig. f10).

A.6 Lemma A.2 (finite-draw bias directions, E9)

With m Bernoulli draws per item, $\hat{p}_i \sim m^{-1}\mathrm{Bin}(m, p_i)$: (i) by Lemma A.0's concavity and Jensen, $E[d(\hat{p}_i)] \le d(p_i)$ pointwise (smoothing bias down); (ii) the top-k selection over noisy capacities satisfies E[max-k mean of $d(\hat{p})$] ≥ max-k mean of $E[d(\hat{p})]$ (selection bias up). The net sign is regime-dependent; measured: −40% (m = 2) to +4% (m = 10) to <1% (m = 50), motivating the m ≥ 10 prescription and the simple-bound fallback, which depends on $\hat{p}$ only through the unbiased grand mean. [`p1_t3_finite_draws.csv`; fig. f11]

A.7 Lemma A.3 (Imbens–Manski critical value)

The equation $\Phi(c + w) - \Phi(-c) = 1 - \alpha$ has a unique root $c(w) \in$ [z_{1−α}, z_{1−α/2}] for every $w \ge 0$; c is continuous and nonincreasing in w with c(0) = z_{1−α/2} and c(∞) = z_{1−α}. *Proof.* The left side is strictly increasing in c and increasing in w; evaluate at the bracket endpoints. □ Hence the parameter-level interval [θ̂_lo − c·σ̂_lo, θ̂_hi + c·σ̂_hi] has asymptotic coverage ≥ 1 − α for every θ in the identified set, with equality at the endpoints (Imbens & Manski, 2004, Thm 1 conditions: endpoint estimators jointly asymptotically normal — supplied here by Theorem 4's machinery — and identified-set width consistently estimated). [`test_im_critical_value_limits`, `test_im_worst_case_parameter_coverage`]

A.8 Spillover-robust interval (U4)

If additionally |n⁻¹Σ_{i:cᵢ=0}(yᵢ − y*ᵢ)| ≤ ε (bounded net drift on clean items), every bound above extends by widening both endpoints by ε; the proof of Prop. 1 goes through with the decomposition $\theta = \hat{\mu}$ − (contaminated deflation) − (clean-item drift). ε is estimable from twin experiments (measured 0.003–0.30, growing with dose). [`test_twosided_upper_moves_only_with_lam_minus`; `p2b_spillover_fix.csv`]

A.9 Theorem 5 (minimax optimality of the simple bound in the binary regime)

Setting. Single-draw binary regime R2 under the iid item sampling of Theorem 4: items are drawn from a population P over (p^*, c, δ); the analyst observes one draw $y \sim \mathrm{Bernoulli}(p)$ per item, $p = p^* + c\delta$, with

$E[c] \le \pi$ and, in CSM±, $-\Lambda^{-}p^* \le \delta \le \Lambda^{+}(1 - p^*)$. The estimand is $\theta(P) = E[p^*]$; write $\mu(P) = E[p]$ and $\Lambda = \Lambda^{+}$.

Theorem 5 (statement). (i) With one draw per item the observable law is iid Bernoulli($\mu(P)$): populations with equal μ are exactly indistinguishable at every sample size. (ii) For interior μ ($\pi\Lambda \le \mu \le 1 - \pi(1-\Lambda)$), the identified set for θ given μ is exactly $[\mu - \pi\Lambda, \mu]$ at $\rho = 0$, and the width $\pi\Lambda$ is attained by a feasible population. (iii) Any interval procedure uniformly valid over the model class has width at least $\pi\Lambda$; any claim-robustness procedure can certify at most $\Gamma^* = \Delta/(\pi(1+\rho))$. The reduction of §9.2 is therefore not a limitation of our estimator: it is the full information content of single-draw binary data.

Proof. (i) Conditional on an item, y is Bernoulli(p); marginalising over P, a single draw is Bernoulli(E[p]) and items are iid, so the sample is iid Bernoulli(μ). Higher moments of p are not identified with one draw per item. (ii) Upper endpoint: at $\rho = 0$, $\delta \ge 0$, so $\theta \le \mu$, attained by the uncontaminated population $p^* \equiv \mu$. Lower endpoint: feasibility requires $\delta \le \min(p, \Lambda(1 - p^*))$ and the two constraints cross at $p = \Lambda$, where $p^* = 0$ and $\delta = \Lambda$ are simultaneously feasible; so per-item bias is at most Λ and $E[c\delta] \le \pi\Lambda$. Attainment: let P_2 place mass π on contaminated items with $p^* = 0$, $\delta = \Lambda$ (hence $p = \Lambda$) and mass $1 - \pi$ on clean items with $p^* = (\mu - \pi\Lambda)/(1 - \pi)$, which lies in [0, 1] exactly on the stated interior range. Then $\mu(P_2) = \mu$ and $\theta(P_2) = \mu - \pi\Lambda$. The population P_1 with $p^* \equiv \mu$, $c \equiv 0$ has $\mu(P_1) = \mu$ and $\theta(P_1) = \mu$. By (i) the two are indistinguishable, so both endpoints belong to the identified set, and by the bias bound nothing below $\mu - \pi\Lambda$ does. (iii) A uniformly valid interval must contain $\theta(P_1) = \mu$ and $\theta(P_2) = \mu - \pi\Lambda$ on data laws it cannot tell apart, hence has width $\ge \pi\Lambda$. For a claim $A \succ B$ with margin Δ, applying the P_2 construction to model A (deflation at Λ^{+}) and its mirror to model B (inflation at $\Lambda^{-} = \rho\Lambda^{+}$) reverses the population margin whenever $\pi\Lambda(1 + \rho) > \Delta$ while leaving both observable laws unchanged; hence no procedure can certify robustness at any $\Lambda > \Delta/(\pi(1+\rho))$. ■

The library's `max_bias_simple` reports exactly this width, and `gamma_star(..., simple=True)` the certificate ceiling. [`test_minimax_indistinguishable_pair`, `test_minimax_no_narrower_certificate`]

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